Network Science
CLL798D / CHL7903

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Project Abstract: Hierarchical Topology of Chemical Space – Reproducing the ClassyFire/ChemOnt Classification Taxonomy

Perspective

Chemical classification underpins how chemical engineers and materials scientists navigate the enormous space of known compounds – grouping molecules by shared scaffolds and functional groups so that structure–property relationships, process-safety data, and synthesis routes can be organized and searched at scale. ClassyFire is the automated, rule-based classification system that assigns every compound a place in ChemOnt, a purely structure-derived taxonomy, and it has become a standard annotation layer across cheminformatics resources used in chemical process and materials-informatics pipelines. This project addresses Pathway 1: we will first *reproduce*, independently and at full scale, the core quantitative claims of the ClassyFire paper and its open data release – streaming all 73.1 million rows of the raw dataset exactly once, without materializing the decompressed file, to check compound counts, external-identifier coverage, and taxonomy structure against what the paper and the dataset's release notes state. We will then *extend* this baseline with original network-science analysis of the compound-to-taxonomy assignment itself, carried out and presented as a distinct, clearly-labeled later phase.

Objectives

- Verify the dataset's headline scale claim by an independent streaming count of unique compounds, rather than trusting the stated figure.
- Verify external-identifier coverage claims (PubChem CID and ZINC identifier) against the values stated in the Zenodo release.
- Reconstruct the ChemOnt taxonomy as a graph from its published parent–child dictionary and verify the paper's structural claims: a single computable, structure-derived taxonomy of more than 4,800 categories, organized as a strict tree reaching a maximum depth of eleven levels.
- Characterize the taxonomy's own shape – depth distribution, branching factor, and the largest-fan-out categories – as a baseline description of the classification scheme.
- **Extension:** treat the compound-to-taxonomy assignment as a network-science object: (a) exact closed-form topology of the five compound-similarity graphs induced by shared classification at each hierarchy level, avoiding the combinatorial impossibility of materializing graphs whose naive edge count reaches $\sim 10^{15}$; (b) a class-to-class similarity network built from cross-classification ("alternative parent") annotations, tested for small-world structure; (c) community detection via spectral bisection and Louvain modularity optimization, compared against the taxonomy's own kingdom/superclass/class partitions; and (d) a Monte Carlo simulation of how a classification error introduced at one level propagates upward through the hierarchy.

Dataset and Paper Links

Each record maps one compound (InChIKey, SMILES, and, where available, its PubChem CID and ZINC identifier) to its five-level ChemOnt classification path – kingdom, superclass, class, subclass, and direct parent – stored as compact numeric category IDs alongside non-hierarchical labels (e.g. alternative parents, substituents). The release ships as a single tab-separated file together with a ChemOnt dictionary that resolves each numeric ID to its category name and parent, so the taxonomy itself can be reconstructed as a graph independently of the compound records.

- **Dataset**: InChIKey-Deduplicated ClassyFire/ChemOnt Label Collection, v3. Zenodo. https://zenodo.org/records/20472700 (DOI: 10.5281/zenodo.20472700, CC-BY-4.0)
  Compounds: 73,105,281 | ChemOnt categories: 4,825 | With PubChem CID: 68,693,298 | With ZINC identifier: 30,187,323
- **Baseline Paper**: Djoumbou Feunang, Y., Eisner, R., Knox, C., Chepelev, L., Hastings, J., Owen, G., Fahy, E., Steinbeck, C., Subramanian, S., Bolton, E., Greiner, R., and Wishart, D. S. (2016). ClassyFire: automated chemical classification with a comprehensive, computable taxonomy. *Journal of Cheminformatics*, 8, 61.
  DOI: https://doi.org/10.1186/s13321-016-0174-y

Thank You